Trainings

General Information

The STC control valve uses fuel pressure and spring force to control the position of an AFC-style plunger. The position of the plunger dictates whether the oil passage to the hydraulic tappets is open or closed. Fuel pressure acts on the piston end of the plunger.

During advanced timing (low fuel pressure), the spring opposes the fuel pressure and holds the plunger in the open position. Pressurized engine lubricating oil flows to the tappets and initiates advanced engine timing. As fuel pressure increases, the spring holds the plunger in the open position until the fuel pressure rises above the certified switch pressure.

At the certified level, the higher fuel pressure overcomes the spring. The action shifts the plunger and closes the oil passage. The oil supply to the tappets is interrupted, and the engine begins to operate in the normal timing mode.

The oil control valve is calibrated to a specific flow and pressure using a fuel pump test stand. Correct valve operation is necessary to maintain acceptable cylinder pressures and white smoke levels, and to assure optimum fuel economy.

Calibration of the valve is carried out prior to shipment to the field or installation on engines, so they do **not** require recalibration. However, valve calibration can be checked and reset in the field, if necessary, by following the calibration procedure later in this section.

The hydro-mechanical STC valve was released on K38 engines in February, 1988 (engine serial number first 33113896) and on K50 engines in April, 1988 (engine serial number first 33113563).

Color coded Viton™ o-rings were released in January, 1989 (engine serial number first 33115382), to eliminate hardening and leakage. This replaced the buna-nitrile material previously used.

Valves built prior to January, 1990 were built with a diaphragm. When rebuilding these valves, use a gasket and modify the plumbing for appropriate venting.

A revised barrel and plunger was released in January, 1990 for parts standardization. The revised barrel had eight oil passage holes (four inlet and four outlet).

The inlet oil filter screen and screen retainer were obsoleted for production and service in all valve's. The screen and retainer were removed to reduce the valves restriction to oil flow.

A modified valve disc and stainless steel capscrews were released in March, 1991 (engine serial number first 33119440), to eliminate leakage.

An STC valve repair kit and STC valve capscrew kit is available.

The valve repair kit contains springs, diaphragms, gaskets, o-ring seals, etc., so that one kit can be used to repair all STC valve part numbers.

The STC valve capscrew kit contains stainless steel capscrews used to assemble the STC valve. The stainless steel capscrews replaced the plain steel capscrews on all STC valves built for production and service from March, 1991, date code “03-91” to reduce capscrew corrosion and subsequent disassembly problems.

The following table lists calibration specification, Hydro-mechanical STC Oil Control Valve (Mechanical), Calibrate.

Valve Part Number Service	**Rail Drain**	**Spring Part Number**	**CPL**	**Engine Model**	**Upper Shift* Point kPa [psi]**	**Lower Shift* Point kPa [psi]**
3060609	Yes	3051069	816	KTTA50-C	386 [56]	186 [27]
			846	KTTA50-G/GS/GC		
			878	KTA50-C		
			968	KTTA50-C		
			1219	KTA50-G/GS/GC		
			1254	KTTA50-G/GS/GC		
3063022	Yes	3051069	844	KTA38-C	538 [78]	262 [38]
			859	KTA38-C		
			872	KTTA38-C/L		
			876	KTTA38-G/GS/GC		
			1251	KTA38-G		
3076333	Yes	3056072	2256	KTA38-G6/G7	131 [19]	55 [8]
			2257	KTA38-G6/G7		
			6238	KTA50-G6/G7		
3633381	Yes	3056072	2354	KTA-50-GS8	324 [47]	21 [3]
			2589	KTA50-M2		
			8194	KTA50-DM2		

			8688	K50-G8		
			8713	QSK50-G8		
			2706	KTA50-G8		
			2707	KTA50-G8		
			3730	K50-M		
			3763	KTA38-M2		
			3773	QSKTA50-CE		
			5606	KTA38-M2		

⚠ **WARNING** ⚠

Diesel engine exhaust and some of its constituents are known to the state of California to cause cancer, birth defects, and other reproductive harm.

In May, 1997 information on the introduction of the dual STC Valve Upfit Kits was released. This kit enabled the upfit of K38 and K50 engines built after engine serial number first 33125767. Each kit contained parts for both left and right banks.

The dual STC retrofit was compatible with engines that were built after October, 1993 (engine serial number first 33125767), and equipped with diagnostic hardware.

The retrofit kits were released to support field conversions to the dual STC valve configurations. The dual STC configuration, along with larger hoses, reduces white smoke in cold climates by lowering flow restriction in the oil hoses.

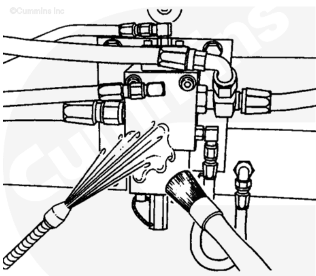
In order to install the dual STC kit on engines built prior to engine serial number first 33125767, the appropriate diagnostic fuel block kit **must** be installed.

Dual STC was released as standard equipment on K2000E, K1800E, K1500E, K38M2, K50M2, and KTA50C engines from April, 1997.

Preparatory Steps

⚠ **WARNING** ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



06600214

⚠ **WARNING** ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the STC oil control valve and the surrounding area before removing it from the engine.

Use solvent to clean the valve.

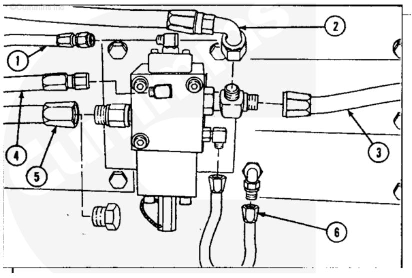
Dry with compressed air.

Remove

Remove the following lines from the STC oil control valve.

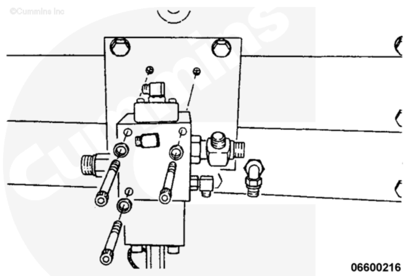
Plug the end of the oil supply line to prevent the oil rifle from draining.

- Fuel pressure sensing line (1)
- Oil outlet lines to tappets (2) and (3)
- Fuel leakage return line (4)
- Oil supply line (5)
- Oil drain line (6)



Note : Depending on the STC valve part number, the valve's have different combinations of plumbing for locations 4, 5, and 6.

Remove the three capscrews that hold the STC oil control valve to the mounting bracket. Remove the STC control valve.

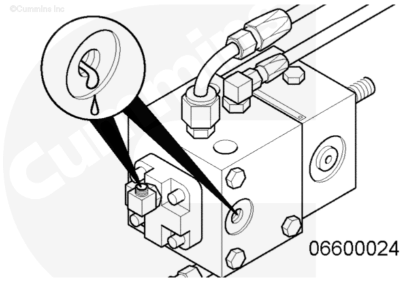


Clean and Inspect for Reuse

Examine the valve for any signs of cracking or leakage.

If leakage is suspected, go to the test section of this procedure.

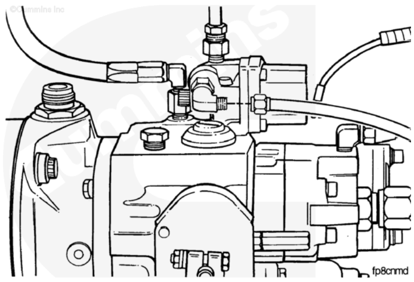
If cracks are detected, the valve **must** be replaced.



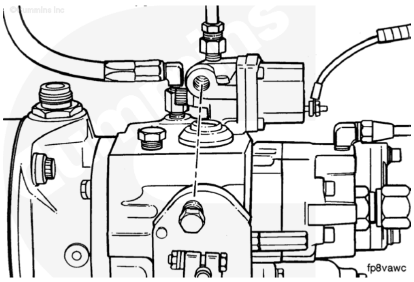
Test

Single STC

Disconnect the STC fuel pressure sensing line from the fuel pump shutoff valve.



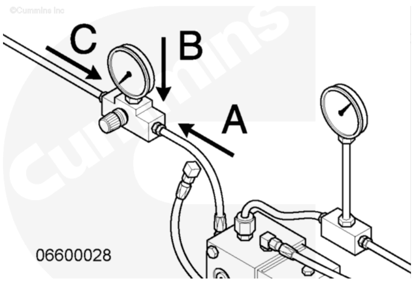
Use a pipe plug to plug the hole in the shutoff valve.



Connect the STC fuel pressure line (A) to the outlet side of the air pressure regulator, Part Number ST-990-40.

Connect a 0 to 693 kPa [0 to 100 psi] pressure gauge (B), Part Number 3375275, to the air pressure regulator.

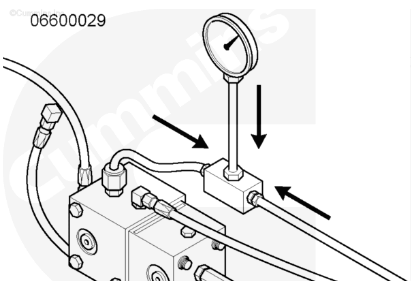
Connect a shop air pressure line (C) capable of providing 693 kPa [100 psi] to the inlet side of the regulator.



Disconnect the STC oil outlet line from the STC oil control valve.

Connect a tee fitting, Part Number ST434-1, between the oil outlet fitting and oil outlet line.

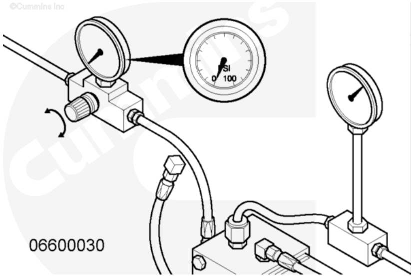
Connect a 0 to 418 kPa [0 to 60 psi] pressure gauge, Part Number 3823204, to the tee fitting.



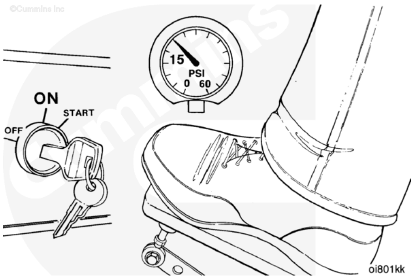
Verify the fuel leakage line check valve is installed.

Failure to have the fuel leakage line check valve installed will result in inaccurate low shift point measurements.

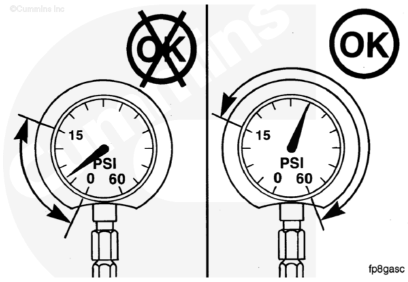
Adjust the air pressure regulator until the air pressure gauge shows 0 kPa [0 psi].



Operate the engine at idle, or at a speed great enough to maintain at least 103 kPa [15 psi] oil pressure between the STC oil control valve and the tappets.

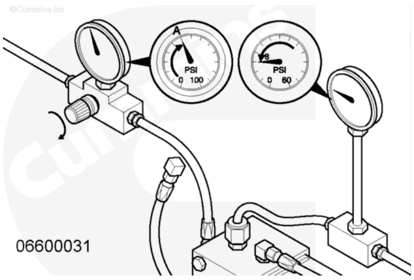


If the oil pressure will **not** reach 103 kPa [15 psi], even at high idle, the STC valve is **not** functioning properly, the oil supply line or check valve is restricted, or the oil supply system is **not** functioning properly. See Section 7 to troubleshoot the oil supply system.



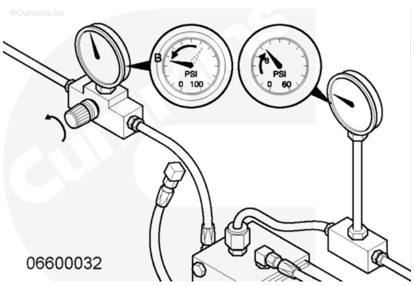
Slowly adjust the pressure regulator to increase the air pressure, until the oil pressure drops below 55 kPa [8 psi]. Record the air pressure at this point. This is the upper shift-point (A).

Note : The sound of the overhead will become quieter when the 55 to 69 kPa [8 to 10 psi] point is reached. This is the shift from advanced to normal timing.



Raise the air pressure at least 103 kPa [15 psi] above the upper shift-point, then slowly begin to lower it. Record the air pressure at which the oil pressure raises above 55 kPa [8 psi]. This is the lower shift-point (B).

Note : The lower shift-point is the shift from normal to advanced timing. The overhead will go from quiet to



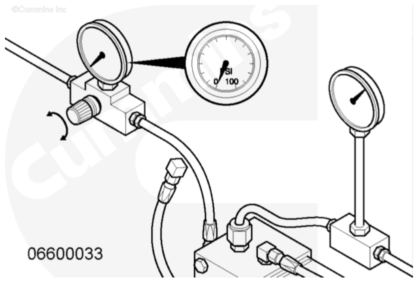
noisy. This pressure must be lower than the upper shift-point. If it is not, rebuild or replace the STC oil control valve.

Compare the shift-point values of those specified for the valve in the Calibrate section of this procedure.

Leak Test

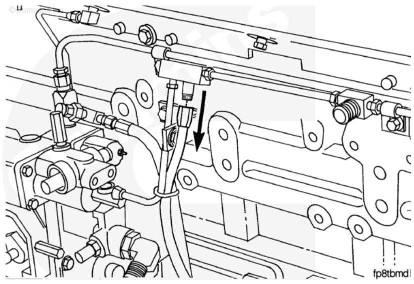
Disc Valve

Adjust the air pressure regulator until the air pressure gauge shows 0 kPa [0 psi].



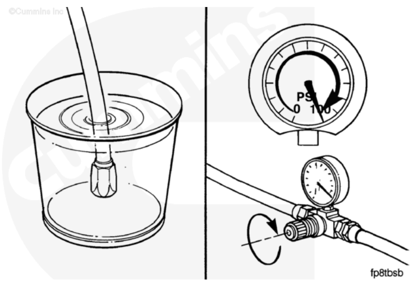
Disconnect the fuel drain line from the injector return line and cap the fitting.

If the STC oil control valve does **not** have a rail drain line, remove the rail drain plug from the STC oil control valve and install a hose approximately 0.5 meters [19.2 in] in length.



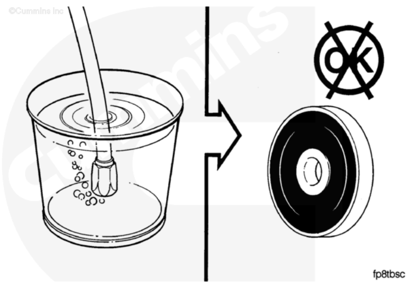
Put the fuel drain line into a bucket of water.

Apply 693 kPa [100 psi] air pressure to the rail inlet.



Observe the bucket for air bubbles.

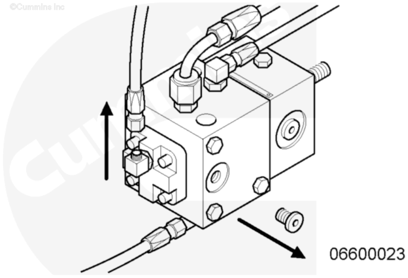
If leakage exists, repair or replace the valve.



O-ring

Disconnect the fuel pressure signal (rail) line.

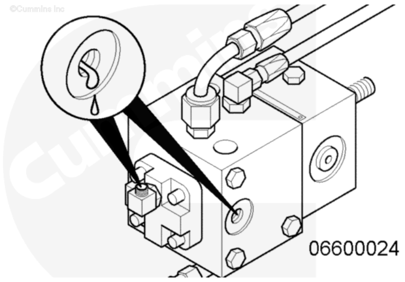
Disconnect the fuel return (rail drain) line or remove the plug.



Inspect the lines for engine lubricating oil.

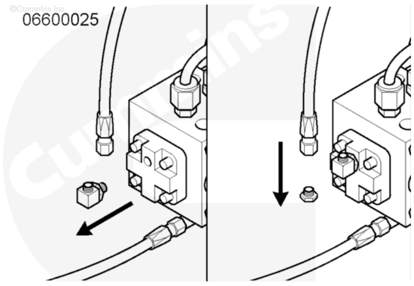
If engine lubricating oil is present, the o-rings will leak. Repair or replace the STC oil control valve.

Note : Some engine operators blend engine lubricating oil into the fuel. Discolored fuel can be expected when blending is practiced. If the customer blends, continue below.



Install plugs into the fuel pressure signal line and the fuel return line (if equipped).

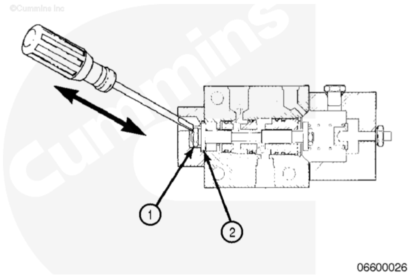
Remove the fuel pressure elbow fitting from the STC oil control valve.



Start and operate the engine at 1200 RPM.

Use a small screwdriver to manually shift the STC oil control valve several times by applying pressure to the disc valve retaining nut.

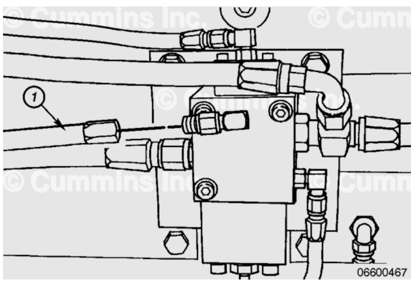
If oil is observed, repair or replace the STC oil control valve.



Fuel Return

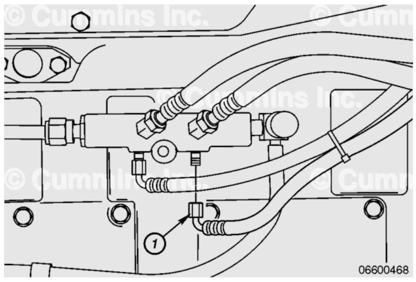
Disconnect the fuel return hose from the STC oil control valve (1).

Remove the fuel return elbow from the STC oil control valve and plug the hole with a 3/8-24 UNF plug.



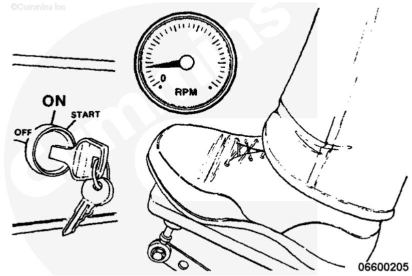
Disconnect the fuel return hose from the fuel block (1).

Remove the nipple from the fuel block and plug the hole with a 1/8 NPTF plug.



Operate the engine under normal operating conditions. Shut down the engine. If the engine does **not** shut down in the recommended time span, replace the STC fuel return elbow.

Install the hoses and repeat the test. If the engine continues to run-on after changing the elbow, change the STC oil control valve and repeat the test.

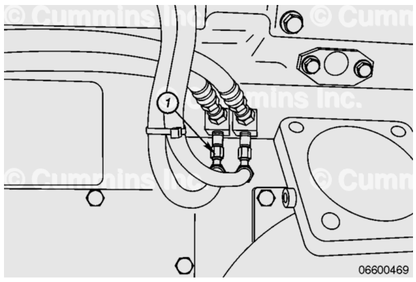


Fuel Return with Dual STC

If the engine is fitted with dual STC, perform the Fuel Return Test on each bank individually to determine which bank leaks.

For right bank inspection, disconnect the fuel return line at the location indicated (1). Remove the nipple and plug the hole.

The right bank fuel return line is situated in the same location on the STC oil control valve as on the left bank STC oil control valve.

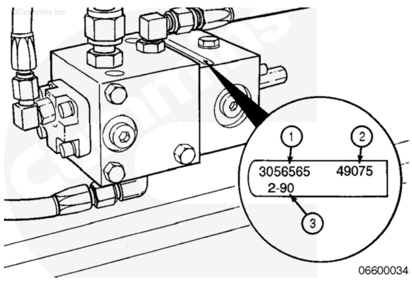


Calibrate

Record the part number of the STC oil control valve from the dataplate on the valve.

Record the upper and lower shift point values for the STC oil control valve.

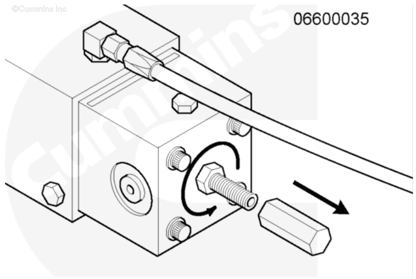
Use the calibration table in STC Oil Control Valve (Mechanical) - General Information.



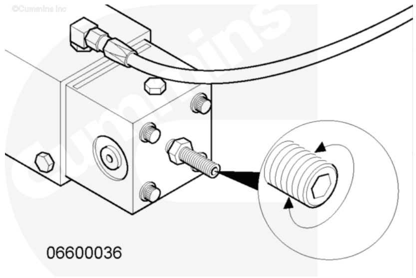
Remove the tamper proof wire from the acorn nut at the rear of the STC oil control valve.

Remove the acorn nut from the STC oil control valve adjusting screw.

Loosen the locknut on the STC oil control valve adjusting screw.

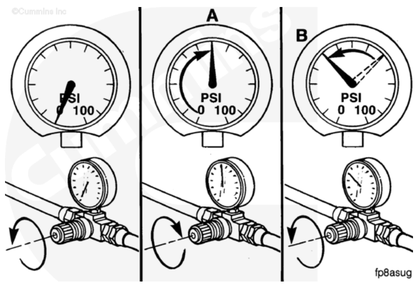


Use an Allen wrench to turn the STC valve adjusting screw. Turn the screw in a **clockwise** direction to raise the STC oil control valve upper shift-point. Turn the screw in a **counterclockwise** direction to lower the upper shift-point.



Note : Always start the air pressure at 0 kPa [0 psi] before checking the upper shift point (A). Always raise the air pressure to at least 103 kPa [15 psi] above the upper shift-point (A) before checking the lower shift-point (B).

After adjusting the screw, repeat the STC shift-point test procedure.



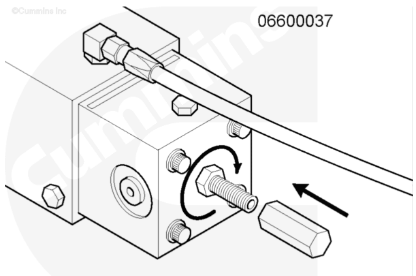
Repeat the calibration and test procedure until the shift-points are within the set values as per the calibration table in the General Information section.

If the value can **not** be calibrated to these specifications, repair or replace the STC oil control valve.

Tighten the STC adjusting screw locknut.

Install the acorn nut on the STC adjusting screw.

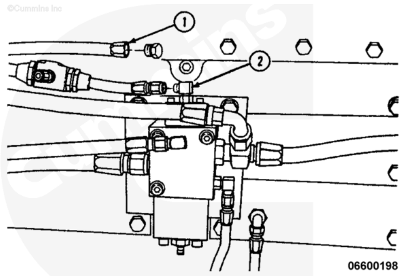
Install the tamper proof wire between the acorn nut and the capscrew.



Dual STC

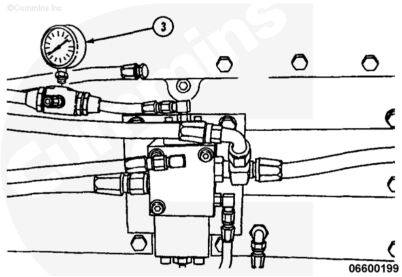
Disconnect fuel pressure signal line (1) and install a plug in it.

Install a regulated air supply to the fuel pressure fitting (2). Use Regulator, Part Number ST-990-40, or equivalent.



Install gauge (3), Part Number 3375275, or equivalent to the air pressure regulator. 0 to 693 kPa [0 to 100 psi].

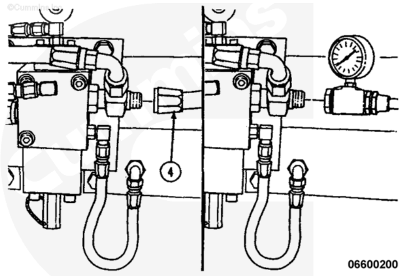
Connect a shop air supply line to the regulator inlet. This procedure requires a 693 kPa [100 psi] air supply.



Remove an oil outlet line (4).

Install a tee fitting, Part Number ST-434-1 and Gauge, Part Number 3823204, or equivalent.

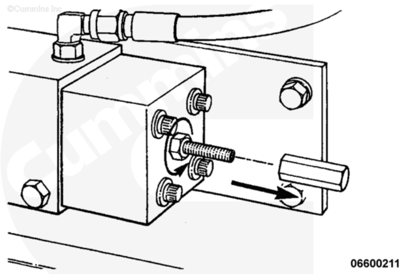
The gauge **must** be capable of 0 to 413 kPa [0 to 60 psi].



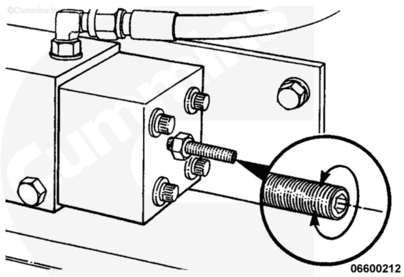
Remove the tamper resistant wire from the acorn nut at the rear of the STC oil control valve.

Remove the acorn nut from the STC oil control valve adjusting screw.

Loosen the lock nut on the STC oil control valve adjusting screw.

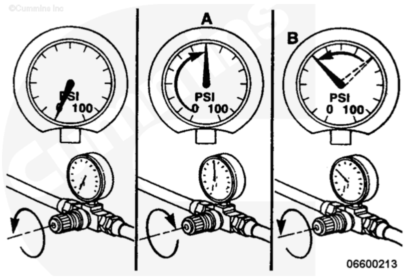


Use an Allen wrench to turn the STC oil control valve adjusting screw. Turn the screw in a **clockwise** direction to raise the STC oil control valve upper shift-point. Turn the screw in a **counterclockwise** direction to lower the upper shift-point.



After adjusting the screw, repeat the STC shift-point check procedure steps.

Note : Always start with the air pressure at 0 psi before checking the upper shift-point (A). Always raise the air pressure to at least 138 kPa [20 psi] above the upper shift point (A) before checking the lower shift-point (8). If the air pressure is not controlled correctly, inaccurate measurements will result.



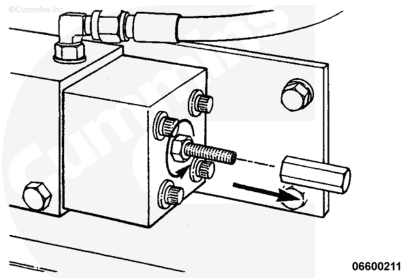
Repeat the calibration and check procedure until the shift points are within the set values listed for each engine model. If the valve can **not** be calibrated to these specifications, replace the STC oil control valve.

Engine Model	**Upper Shift Point Specification**
K38	386 ± 34 kPa [56 ± 5 psi]
K50	538 ± 34 kPa [78 ± 5 psi]

Tighten the STC adjusting screw locknut.

Install the acorn nut on the STC adjusting screw.

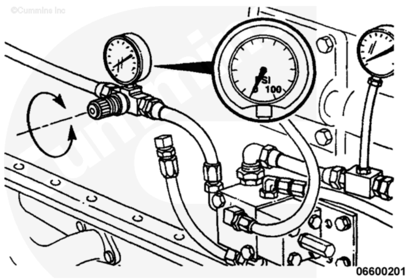
Install the tamper resistant wire between the acorn nut and the capscrew.



Adjust

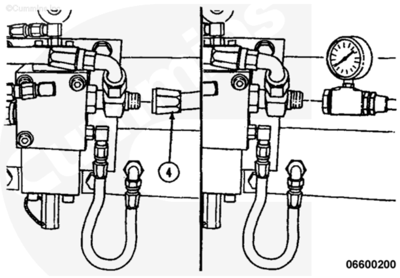
Low Power Complaint

Adjust the air pressure regulator to "0" kPa ["0" psi].



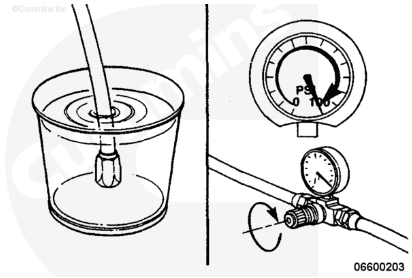
Remove the fuel drain line from the fuel connection block.

Install a cap on the connection block fitting (5).



Place the fuel drain line into a clean container of fuel.

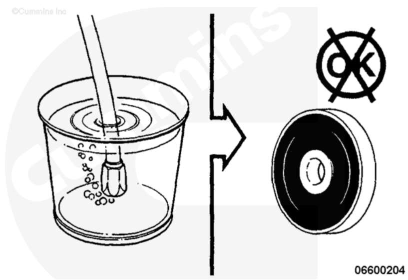
Apply 693 kPa [100 psi] air pressure to the rail inlet line.



Observe the container for air bubbles.

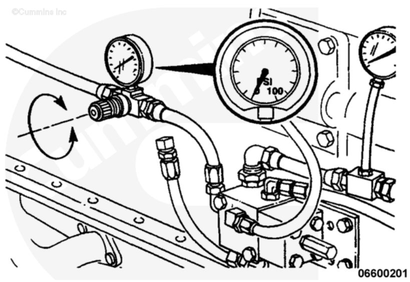
If bubbles are present, the internal disc valve is leaking.

Replace the STC oil control valve.



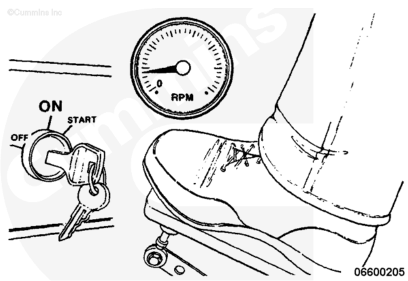
All Other Complaints

Adjust the air pressure regulator until the air pressure gauge show "0" psi.



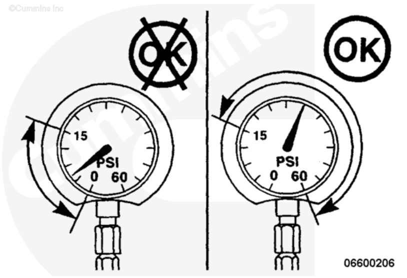
Start the engine and operate at low idle.

Note : The engine speed must be below 800 rpm.



If the oil pressure will **not** reach 103 kPa [15 psi], the STC system is **not** functioning properly. Check the orientation of the oil supply check valve and for restrictions in the oil supply lines. Also check for oil leaks at the connections in the rocker housings.

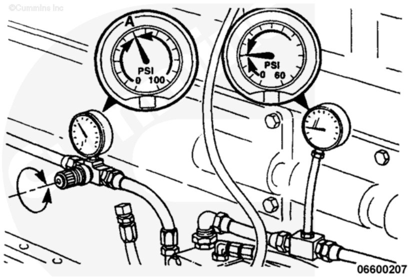
If no fault is found, turn the adjusting screw on the STC oil control valve 10 turns **counterclockwise**. If the pressure is still less than 103 kPa [15 psi], replace the STC oil control valve.



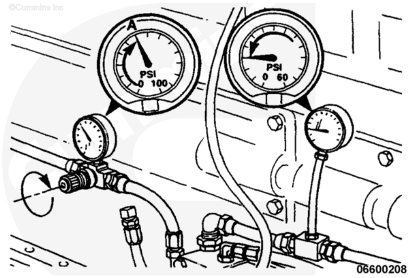
The air pressure on the inlet fitting simulates fuel pressure and is used to cycle the valve between advanced and normal timing. The overhead sounds noisy when in advanced timing, but is much quieter when operating in normal timing.

Note : The decline in oil pressure quiets the overhead.

Cycle the STC oil control valve between advanced and normal timing (using the pressure regulator) to become familiar with the difference in sound of the two timing modes.



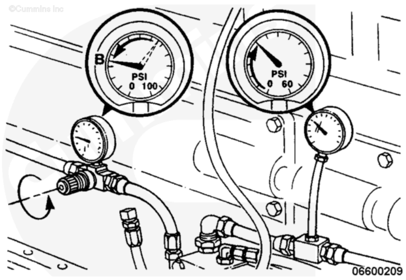
With the air pressure at 0 psi, adjust the pressure regulator to slowly increase the air pressure. Record the air pressure at which the engine overhead sound switches from noisy to quiet (injection timing cycles from advanced to normal). This is called the upper shift point (A).



Raise the air pressure at least 103 kPa [20 psi] above the upper shift point, then slowly begin the lower it. Record the air pressure at which the oil pressure raises and the overhead noise increases.

This is the lower shift point (B). Transition to advanced timing has occurred.

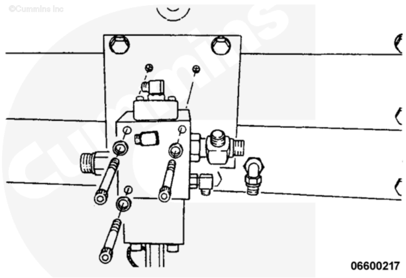
Note : This pressure must be lower than the upper shift point (A). If not, replace the STC oil control valve.



Compare the upper shift-point valves to those specified for the engine model. If the values are **not** within the check specifications listed, calibrate the STC oil control valve.

Engine Model	Upper Shift Point Specification
K38	386 ± 34 kPa [56 ± 5 psi]
K50	538 ± 34 kPa [78 ± 5 psi]

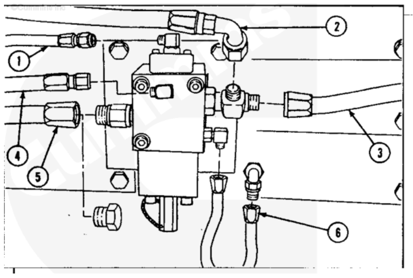
Install



Install the STC oil control valve to the mounting bracket with the three capscrews.

Tighten the capscrews.

Torque Value: 27 n•m [239 in-lb]



Remove the plug from the STC oil supply line (5).

Connect the following lines to the STC oil control valve.

- Fuel pressure sensing line (1)
- Oil outlet lines to tappets (2) and (3)
- Fuel leakage return line (4)
- Oil supply line (5)
- Oil drain line (6)

Tighten the hoses.

Hoses 1,4,6

Torque Value: 7 n•m [62 in-lb]

Hoses 2,3,5

Torque Value: 27 n•m [239 in-lb]

Finishing Steps

- Operate the engine and check for leaks.

Last Modified: 01-Oct-2025